

Generally, a recession that is associated with a financial crisis lasts longer. It leads to a greater decrease in output and employment; one, directly through the collapse of the financial sector and second, through the consequential economic downturn. Financial crises also have a greater implication for fiscal outlays as governments have to focus on saving the financial institutions along with improving the aggregate demand in the economy.

24.6 POLICY RESPONSES TO A CRISIS

Now, that we know the adverse impacts of a financial crisis or an economic crisis the question arises what can be done? In this section, we will discuss in detail the macroeconomic policies to tackle a crisis. A single policy intervention, like monetary or fiscal, or both, is generally not sufficient for stability in the economy. A combination of different kinds of policy measures is required. The following policy measures can be used to fight a crisis.

Fiscal Policy: We know that in a crisis, financial or economic, unemployment increases, and output and aggregate demand decrease. Policymakers can use fiscal expansion, in the form of a tax cut or an increase in government spending, to increase aggregate demand in the economy. A tax cut would lead to an increase in disposable income in the economy, while, an increase in government spending can come in the form of expenditure on infrastructure, stimulus to industry, social infrastructure (health and education), and social welfare schemes. The magnitude of fiscal intervention that is required to counter the cyclical downturn depends upon the seriousness of the crisis. The bigger the crisis, the larger should be the fiscal expansion.

Monetary Policy: Like fiscal expansion, monetary policy can also be used to increase aggregate demand in the economy which would help in improving employment and output. A monetary expansion generally comes in the form of a reduction in interest rates and an increase in the liquidity or money supply in the economy. A decrease in interest rates would lead to a reduction in the cost of borrowing, and an increase in liquidity would help in easing the credit crunch in the economy.

Both monetary and fiscal policies have their limitations. A deep crisis may require a drastic decrease in the rate of interest but it cannot go below zero. Moreover, monetary policy is not useful in a condition of a liquidity trap. Fiscal expansion may hurt the interest rates and prices in the economy. In such situations, monetary and fiscal policy should be supplemented by different regulations directly addressing the issues related to asymmetric information, adverse selection, and moral hazard.

Strengthening of Regulatory Framework: Financial crisis is generally a result of asset-price bubbles. Policymakers should strengthen the regulatory and monitoring process of the financial sector. This requires continuous

supervision and evaluation of financial institutions and financial instruments. Financial innovations take place continuously in the market, in the form of new products and expansion of the market, which creates new challenges not only for the financial institutions and the markets but also the policymakers and regulators. This creates a need for continuous improvement in the regulatory framework.

Check Your Progress 2

- 1) Explain asset-price boom and bust with a suitable example.

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- 2) What do you understand by fire sale?

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- 3) How do financial and economic factors interact during a crisis?

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- 4) What policy options can be used to fight a recession?

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24.7 LET US SUM UP

The financial sector is very important for the overall growth of the economy. The major function of this sector is to provide channels for the movement of funds between borrowers and lenders. Speculative activities in the financial sector have major implications for the business cycles. The recession of 2007-08 better known as the subprime mortgage crisis, provides a very good example of this. Over the years many financial innovations have taken place which has helped not only in building capital base but also in expansion of

markets and wealth across the countries. However, the process or path of economic growth is not smooth or stable. Economies of different countries have experienced wide fluctuations in variables like employment, output, and prices over the past three centuries. The presence of asymmetric information, adverse selection better known as the subprime mortgage crisis, and moral hazard can also contribute to the crises. As financial development helps in economic growth a financial crisis worsens the economic problems. Booms in asset prices, fuelled by credit growth and positive expectations, followed by busts are generally the cause of the financial crisis. It is not necessary that a financial crisis would lead to an economic crisis; an economic crisis may also be a cause for the financial crisis. Finance is a key factor in capital formation and business activities. The policy response can be in terms of fiscal policy, monetary policy and regulatory reforms.

24.8 KEYWORDS

Business Cycle	: Refers to ups and downs in economic activities namely, output, unemployment, and income among other things.
Real Economy	: The real economy produces goods and services with the help of the factors of production, land, labour, and capital. It provides us with consumption goods like food, furniture, clothes, and capital goods like plant, machinery, transportation etc.
Asymmetric Information:	When in a transaction, one party does not have the same information as the other party it is known as asymmetric information.
Adverse Selection	: When the lenders/investors, pursued by the borrower, invest in the project without knowing the risk.
Moral Hazard	: This takes place after the financial transaction has occurred. A borrower may divert the borrowed funds to some risky project or activity.
Credit Boom	: With the easy availability of credit, investors may increase their investment in an asset they expect to give very good returns in the future.
Monetary Policy	: Increase or decrease in money supply by the central bank to target macroeconomic variables like unemployment, output, inflation etc.
Fiscal Policy	: Used for increase in output through Government purchases and taxes.

24.9 SOME USEFUL REFERENCES

Burton, Maureen., Nesiba, Reynold F., Brown, Bruce. *An Introduction to Financial Markets and Institutions*. United Kingdom: Taylor & Francis, 2015.

Howells, Peter., Bain, Keith. *Financial Markets and Institutions*. Germany: Pearson Education, 2007.

Mankiw, N. Gregory. *Macroeconomics*. United States: Worth Publishers, 2016.

Mishkin, F.S., and S.G. Eakins. *Financial Markets and Institutions*. Pearson Education Series in Finance. Prentice Hall, 2012.

Mishkin, F.S. *Macroeconomics: Policy and Practice*. Macroeconomics: Policy and Practice. Pearson Education, 2012.

Further Readings:

Cecchetti, Stephen, Marion Kohler, and Christian Upper. "Financial Crises and Economic Activity." Cambridge, MA: National Bureau of Economic Research, September 2009. <https://doi.org/10.3386/w15379>.

Claessens, Stijn, and M Ayhan Kose. "Financial Crises: Explanations, Types, and Implications," IMF Working Paper No. 2013/028

24.10 ANSWERS / HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Economic expansion, economic activities are at their peak, and output starts to slow down and then downfall.
- 2) Refer to Sub-section 24.3.2
- 3) credit or default risk, liquidity risk, interest rate and price risk, and exchange rate risk.

Check Your Progress 2

- 1) At times, the prices of financial assets like stocks or real estate properties increase by a large amount over their **fundamental values**. This is known as **the asset-price bubble**.
- 2) The selling of assets by the banks may lead to a rapid decline in asset prices. This is also known as **a fire sale**.
- 3) Refer to Section 24.5
- 4) Fiscal Policy, Monetary Policy, Strengthening of Regulatory Framework

UNIT 25 EFFICIENT MARKET HYPOTHESIS

Structure

- 25.0 Objectives
- 25.1 Introduction
- 25.2 History of Efficient Market Hypothesis (EMH)
- 25.3 Efficient Market Hypothesis
- 25.4 Assumptions of EMH
- 25.5 EMH and Capital Asset Pricing Model (CAPM)
- 25.6 Assessment of Efficient Markets Hypothesis
- 25.7 Applications of the EMH
- 25.8 Applicability of the EMH in India
- 25.9 Let Us Sum Up
- 25.10 Key Words
- 25.11 Some Useful Books
- 25.12 Answers/ Hints to Check Your Progress Exercises

25.0 OBJECTIVES

After going through this Unit, you will be able to:

- explain the meaning of the Efficient Markets Hypothesis;
- describe the relationship of the Efficient Markets Hypothesis and Capital Asset Pricing Model (CAPM);
- critically examine the Efficient Markets Hypothesis;
- appreciate empirical techniques to test the Efficient Markets Hypothesis;
- apply the Efficient Markets Hypothesis to real-life data; and
- apply the Efficient Markets Hypothesis to data on Indian financial markets.

25.1 INTRODUCTION

A key idea in contemporary finance, the Efficient Markets Hypothesis (EMH) has revolutionised the way we think about the behaviour of financial markets. According to the hypothesis, financial markets are incredibly effective, and the prices of securities exchanged there always represent all

available information. To put it another way, market prices react quickly and precisely to any new information that enters the market, making it nearly hard for investors to continuously generate above-average profits by trading on public information.

The concept of EMH has been around for more than 50 years now. The hypothesis has been widely debated by scholars and practitioners due to its assumptions. Some scholars argue that EMH is flawed, while others maintain that it provides a realistic and accurate representation of the functioning of financial markets. Nevertheless, EMH has played a crucial role in shaping modern financial theory, and it remains a key subject of study for students and professionals in the field of finance.

The present unit aims to provide a comprehensive introduction to the Efficient Markets Hypothesis and its applications in the field of finance. We explore the origins and evolution of the theory, examine its key assumptions and implications, and review the empirical evidence supporting and challenging its validity. We also discuss various forms of market efficiency, such as weak, semi-strong, and strong efficiency, and how they relate to different types of information available to investors. Finally, we discuss the practical implications of EMH for investment decision-making, asset pricing, and portfolio management.

Whether you are a finance professional, a student, or simply interested in understanding how financial markets work, this unit will provide you with a solid foundation in the Efficient Markets Hypothesis and its importance in the world of finance.

25.2 HISTORY OF EFFICIENT MARKETS HYPOTHESIS

The Efficient Markets Hypothesis (EMH) is a theory in finance that states that financial markets are “informationally efficient,” meaning that asset prices reflect all available information at any given time. The basic idea behind the EMH is that in an efficient market, investors cannot consistently earn abnormal returns (i.e., returns that are higher than what would be expected given the level of risk they are taking on) by using any available information, because that information is already reflected in the market price.

The history of the efficient markets hypothesis dates back to the early 20th century when economists such as John Bates Clark and Alfred Marshall developed theories about the relationship between prices and information. However, the modern version of the EMH began to take shape in the 1950s and 1960s, when economists such as Eugene Fama, Paul Samuelson, and Burton Malkiel began to develop and refine the theory.

Eugene Fama is often credited with creating the modern version of the EMH in his 1965 paper “Random Walks in Stock Market Prices,” which argued

that stock prices follow a random walk and that it is impossible to consistently predict future stock prices based on past prices or any other available information. This paper, along with others by Fama and other economists, helped establish the EMH as a cornerstone of modern finance theory.

Since its inception, the EMH has been the subject of much debate and controversy. Some economists and investors argue that the EMH is an accurate representation of how financial markets work, while others believe that markets are not completely efficient and that it is possible to consistently earn abnormal returns through various investment strategies. Nonetheless, the EMH remains an important concept in finance and continues to influence the way that investors think about the relationship between information and asset prices.

25.3 EFFICIENT MARKETS HYPOTHESIS (EMH)

The Efficient Markets Hypothesis (EMH) is a theory in finance that suggests that financial markets are “efficient,” which means that asset prices fully reflect all available information at any given time. The hypothesis states that it is impossible to consistently achieve returns over average market returns because any new information that enters the market is immediately and fully reflected in the asset prices. In other words, the EMH suggests that it is impossible to consistently beat the market by using information that is already publicly available since the market has already used that information to arrive at the prevailing price. The theory argues that market players cannot earn higher returns than the market as a whole, except by taking on more risk. Eugene Fama (1970) has given three forms of the EMH which are as follows:

- 1) **Weak Form:** The weak form of EMH asserts that all historical price and volume data is already incorporated into current stock prices, making it impossible to profit from analyzing past trends or patterns. This means that technical analysis, which involves using charts and past market data to predict future prices, cannot generate consistent profits.
- 2) **Semi-Strong Form:** The semi-strong form of EMH asserts that all publicly available information is already reflected in stock prices, making it impossible to achieve abnormal returns through fundamental analysis. This means that researching and analyzing company financial statements, news articles, and economic data cannot lead to consistent profits.
- 3) **Strong Form:** The strong form of EMH asserts that all information, both public and private, is already reflected in stock prices, making it impossible to achieve abnormal returns through any form of analysis or insider information.

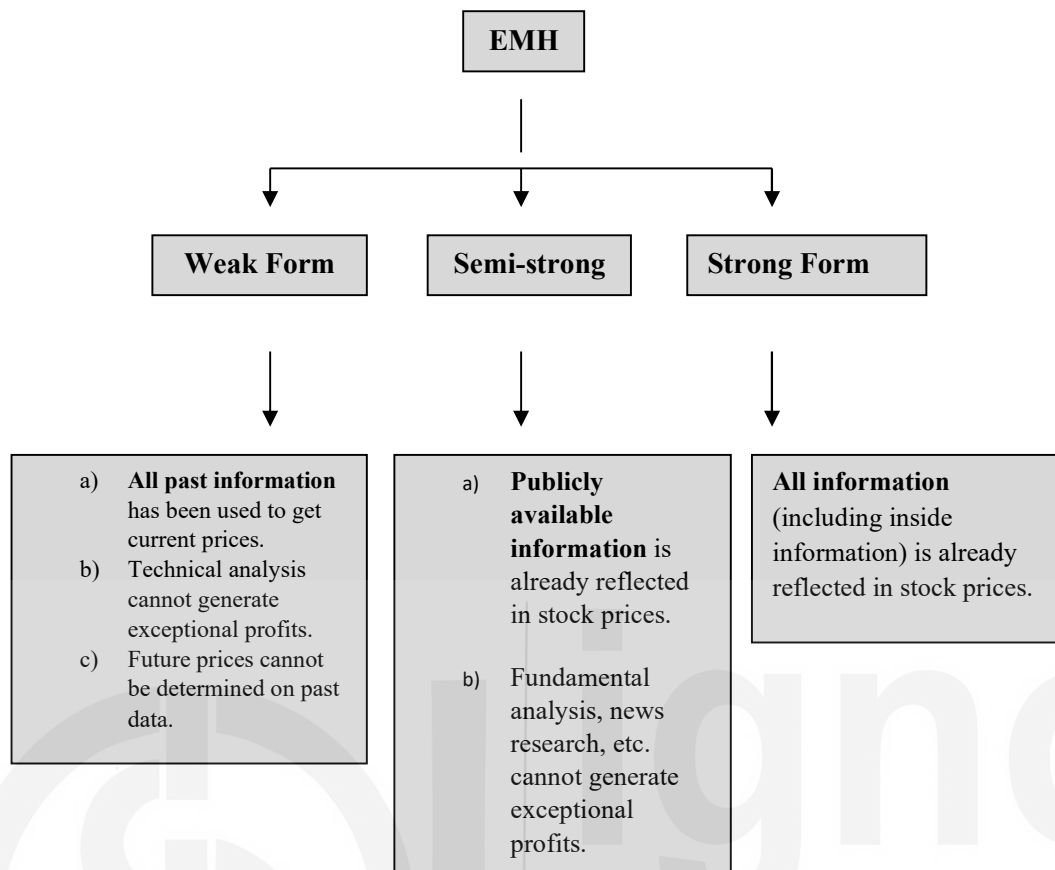


Fig. 25.1: Different Forms of Efficient Markets Hypothesis

While the efficient markets hypothesis is widely debated, it has influenced much of modern financial theory and has important implications for investors, financial analysts, and policymakers.

25.4 ASSUMPTIONS OF EMH

The Efficient Markets Hypothesis is based on the following assumptions:

- 1) **Rationality:** Investors are rational in the sense that they maximise their interests. They do their best.
- 2) **Information is freely available:** Every investor has easy accessibility to all market-related information without any restrictions. In other words, the information is available at zero cost.
- 3) **No asymmetry of information:** It means every investor has the same amount of information i.e., whatever information investor A has, then the same information is available to B also.
- 4) **Prices adjust quickly to new information:** The response time of the market to new information is much less. In other words, as new information is made available, then its effect can be seen in stock prices and indices very quickly.

- 5) **No individual investor can influence prices:** No investor, or group of investors, has the force to control prices and steer the market in a specific direction.
- 6) **No transaction costs:** Investors are not prevented from acting on their information by transaction costs or other trading obstacles.

Check Your Progress 1

- 1) What is the basic idea underlying the Efficient Markets Hypothesis?

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- 2) What are the various forms of the EMH? What do they imply?

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- 3) State the assumptions of the Efficient Markets Hypothesis.

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25.5 EMH AND CAPITAL ASSET PRICING MODEL (CAPM)

The Efficient Markets Hypothesis (EMH) and the Capital Asset Pricing Model (CAPM) are two important concepts in finance that are closely related to each other.

The Efficient Markets Hypothesis states that (i) financial markets are efficient, and (ii) all available information is reflected in asset prices. According to the EMH, it is impossible to consistently beat the market because all available information is already reflected in the asset prices. In other words, the market is already in equilibrium and all the stocks are valued at their intrinsic values. Therefore, a stock cannot perform better than the market.

The Capital Asset Pricing Model (CAPM) as we have already discussed in Unit 20 is an asset pricing model which was suggested by economists William F. Sharpe (1964) and Linter (1965). The CAPM states that the expected return of an asset (or a security portfolio or stock) is linearly and positively related to its systematic risk (the part of total risk which is beyond the control of the firm or the investor; it is determined by various economic, social and political factors). The CAPM predicts that the systematic risk of an

asset has the power to explain the expected return of an asset in a linear fashion. Sharpe and Linter established the following equation of the CAPM:

$$E(R_i) = R_f + [E(R_m) - R_f]\beta_i \quad \dots (25.1)$$

where $E(R_i)$ is the expected return of an asset i , R_f represents the rate of return of a risk-free asset (such as T-bills or Government securities), $E(R_m)$ is the expected return of a market portfolio (typically proxied by an index of a stock exchange such as SENSEX or NIFTY), and β_i is a measure of the systematic risk. Apart, the expression $E(R_m) - R_f$ is called market risk-premium.

The CAPM is based on the following main assumptions:

- a) investors are rational,
- b) they diversify their portfolios among various financial assets or instruments, and
- c) the market is efficient.

The point to be noted here is the assumption that “the market is efficient”. Thus, the CAPM states that the EMH holds good.

Specifically, the CAPM assumes that markets are efficient, which means that all relevant information is already incorporated into asset prices. In other words, the CAPM assumes that investors cannot earn higher returns by using publicly available information, which is a core principle of the EMH.

The CAPM is also used to test the efficiency of markets. If the CAPM holds, it would suggest that markets are efficient, because it implies that investors cannot earn higher returns by using publicly available information. However, if the CAPM is not able to explain returns on assets, it may suggest that markets are not efficient.

An Econometric Technique to Test the EMH Using the CAPM

Black, Jensen, and Scholes (1972) suggested a test statistic to test the CAPM. They used the following equation for each security i :

$$R_{it} - R_{ft} = \alpha_i + \beta_i(R_{mt} - R_{ft}) + u_{it} \quad \dots (25.2)$$

The left-hand side of equation (25.2), that is, $(R_{it} - R_{ft})$ is called ‘the excess return of security i over the risk-free return’. The expression $(R_{mt} - R_{ft})$ on the right-hand side is called ‘market risk premium’ or the excess return of the market portfolio over the risk-free return. Thus, the above equation can be re-written as follows:

$$ER_{it} = \alpha_i + \beta_i ER_{mt} + u_{it} \quad \dots (25.3)$$

In equation (25.3) the term ER_{it} indicates excess returns of the i^{th} security and the term ER_{mt} indicates the average expected return in the market. The coefficient β indicates the systematic risk.

Black, Jensen, and Scholes (1972) suggest that if the CAPM is working, then α_i should not be statistically different from zero for all n securities. Thus, their null hypothesis is $\alpha_i = 0$. If the null hypothesis is rejected, that is, if $\alpha_i \neq 0$, then the market is not efficient in the light of the efficient markets hypothesis. If we plot equation (25.3) on graph paper, we obtain the characteristic line. If the EMH holds good, then the characteristic line should pass through the origin as shown in Fig. 25.2.

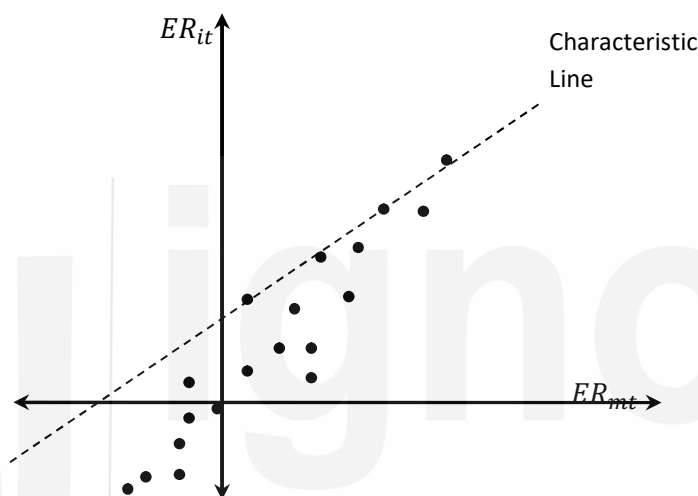


Fig. 25.2: Capital Asset Pricing Model

The overall conclusion that we draw is that the CAPM and the EMH are two closely related concepts in finance. The CAPM is built on the assumptions of the EMH, and the CAPM can be used to test the efficiency of markets.

Check Your Progress 2

- 1) What are the major inferences of the Capital Asset Pricing Model? How is it related to the Efficient Markets Hypothesis?
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- 2) Explain how the EMH can be tested by using the CAPM.
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